

Project Paper Cover Page: Brightness and Brainwaves

Each team member receiving credit for this paper must read the statement below and type their name in the table provided. By entering your name, you acknowledge and agree to the statement.

“I acknowledge that all content in this scientific paper is the result of collaborative work with my designated team members only. We have properly referenced all external material and have adhered to the academic conduct standards outlined in the Lab Syllabus.”

Team Name:	Neuron Ninjas
Team member 1 name:	Ceren Atalar
Team member 2 name:	Beatrice Levine
Team member 3 name:	Eunice Son
Team member 4 name:	Atticus Lu

Results

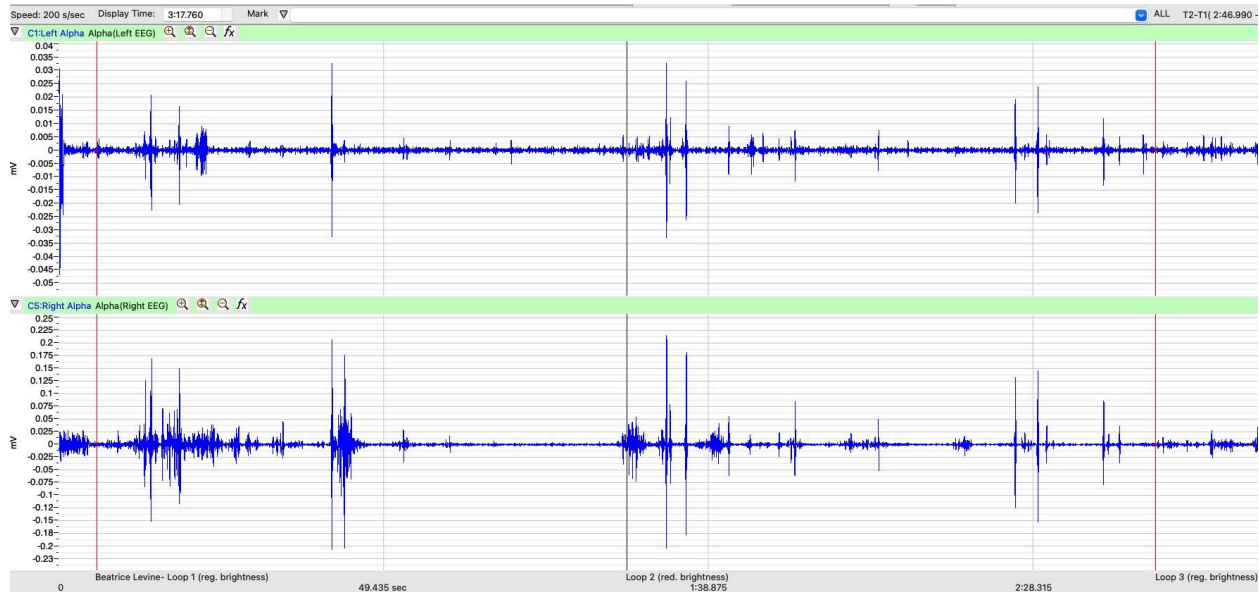


Figure 1 - First two loops of regular and reduced brightness photos for test subject Beatrice Levine. Loops 1 and 2 are visualized between the red cursors to loop at averages across the 4 trials. Similar recordings were found in all other tests with different subjects. This data was taken via ROAM-EEG and IXTA server.

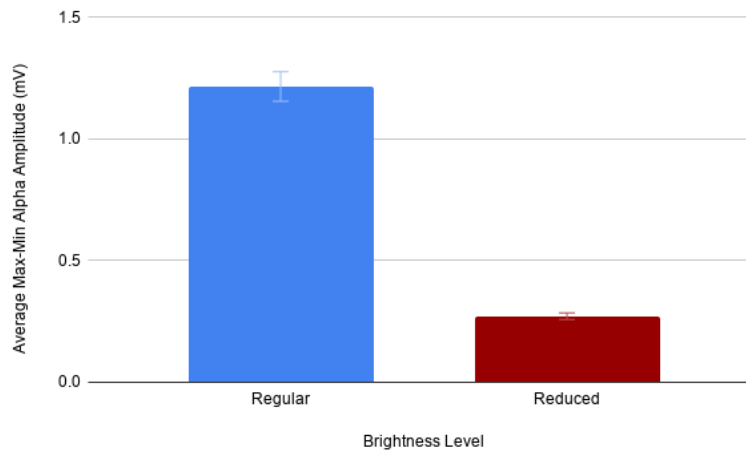


Figure 2 - Average ($\pm 5\%$ Dev) Max-Min Alpha Amplitude (mV) under regular and reduced brightness conditions (all loops $N = 16$). Outliers over 3 SD were not used in the table. Regular brightness conditions include data from Loop 1 and Loop 3, while reduced brightness conditions include Loop 2 and Loop 4.

Average of L/R Cortical Zones (mV)						
	Loop 1 (Regular)	Loop 2 (Reduced)	Loop 3 (Regular)	Loop 4 (Reduced)	Regular Average (Loop 1, 3)	Reduced Average (Loop 2, 4)
Average Max-Min Left Alpha Amplitude (mV)	0.885	0.06	0.098	0.063	0.492	0.062
Average Max-Min Right Alpha Amplitude (mV)	3.473	0.459	0.403	0.501	1.938	0.480
Average Max-Min Alpha Amplitude (mV)	2.179	0.26	0.25	0.282	1.215	0.271

Table 1 - Averaged max-min amplitude values for both left and right cortical zones across multiple experimental loops, as well as their averaged values. This data details the difference in amplitudes for the left and right cortical zones, and quantifies the difference in alpha wave activity upon exposure to regular and reduced brightness visual stimuli.

During our trials, we included four loops with the same 10 images, with two regular brightness trials and two reduced brightness trials. We originally conducted our experiment on 22 test subjects that fit our call for subjects, yet we eliminated the outliers to examine N=16 significant data points, with $df = 15$. Our LS26 Screenshot (Figure 1) demonstrates the first two loops, regular and reduced brightness, for both the Left and Right Alpha EEG from one of our test subjects. Figure 2 is a representation of the average Max-Min left and right amplitudes for the various trials, detailing the difference in amplitudes for each zone, although our data analysis, (Table 1) averages the complete data together as well as the regular and reduced brightness average amplitudes. Overall, a higher brightness is associated with a slightly higher average Max-Min Left and Right Amplitude, indicating cortical arousal, in comparison to reduced

brightness. Brightness level (regular vs. reduced brightness) did not have a statistically significant effect on average alpha amplitude (Figure 2; paired t-test: $t_{15} = 0.0596$; $p = 0.0683$).

Discussion

The objective of the experiment was to investigate the relationship between the brightness of visual stimuli and alpha brain wave activity in the frontal cortex. We hypothesized that increasing the brightness of visual stimuli would significantly decrease the maximum amplitude of alpha brain waves within the frontal cortex. Previous research suggests that alpha waves play an active inhibitory role in visual processing by reducing visual sensitivity in a “gating function” mechanism (Toscani et al. 2010). This suggests that alpha brain activity is strongest when the brain is in a relaxed state, especially when visual input is reduced. However, our findings indicate that brighter stimuli produced larger maximum amplitudes than reduced brightness stimuli (Figure 2), although this difference was not statistically significant. This discrepancy may be explained by the fact that brightness effects on EEG activity are highly dependent on brain region, time window, and stimulus characteristics, rather than producing a uniform decrease in alpha activity (Eroğlu et al. 2020). Environmental conditions such as ambient light and noise may have interfered with our results due to uncontrolled surroundings, conversations, other bright stimuli, and fidgeting. Furthermore, the present study was limited to frontal recordings, whereas visual processing effects are often more pronounced in posterior brain regions. Limitations in our experimental setup may have contributed to the lack of significant findings (Frescura et al. 2023). In order to determine if the brightness of visual stimuli is sufficient in producing significant differences in the recipients’ alpha brain waves, future studies should

adjust the experimental setting to take place within a controlled, dark, and acoustically isolated environment to better isolate the effects of brightness on alpha brain activity.

References

- Eroğlu K, Kayıkçıoğlu T, Osman O (2020) Effect of brightness of visual stimuli on EEG signals. *Behav Brain Res*. 382:112486. doi: 10.1016/j.bbr.2020.112486.
- Toscani M, Marzi T, Righi S, Viggiano MP, Baldassi S (2020) Alpha waves: a neural signature of visual suppression. *Exp Brain Res* 207(3-4):213-9. doi: 10.1007/s00221-010-2444-7.
- Frescura, Alessia & Lee, Pyoung & Jeong, Jeong & Soeta, Yoshiharu. (2023). EEG alpha wave responses to sounds from neighbours in high-rise wood residential buildings. *Building and Environment*. 242. 110560. 10.1016/j.buildenv.2023.110560.